DC Motor



Basic Electrical and Electronics Engineering

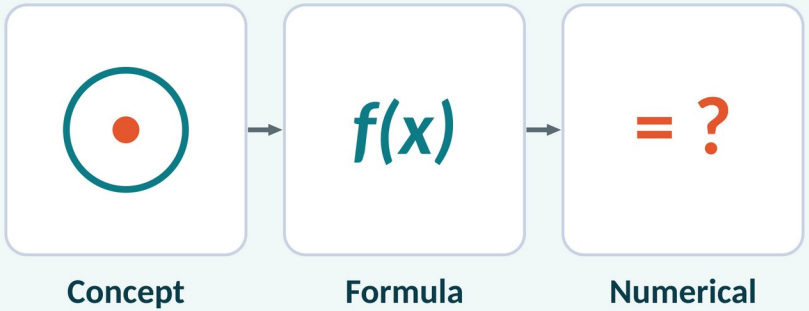
Unit 2: Transformers and Electrical Machines

Topic: Principle, back EMF, torque equation, characteristics and speed control of DC shunt motor

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Learning Outcomes

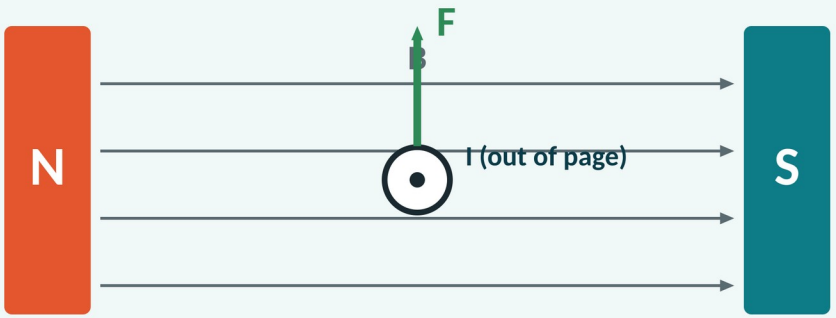
- Explain the principle and parts of a DC motor.
- State the significance of back EMF.
- Use the torque equation and characteristics.
- Control the speed of a shunt motor.



Linked to CO2
Describe construction and operating principle of DC motors.

Recap: What You Already Know

- Magnetic field B around a magnet.
- Force on a current-carrying conductor.
- Fleming's left-hand rule gives direction.
- Faraday: moving conductor in a field induces EMF.



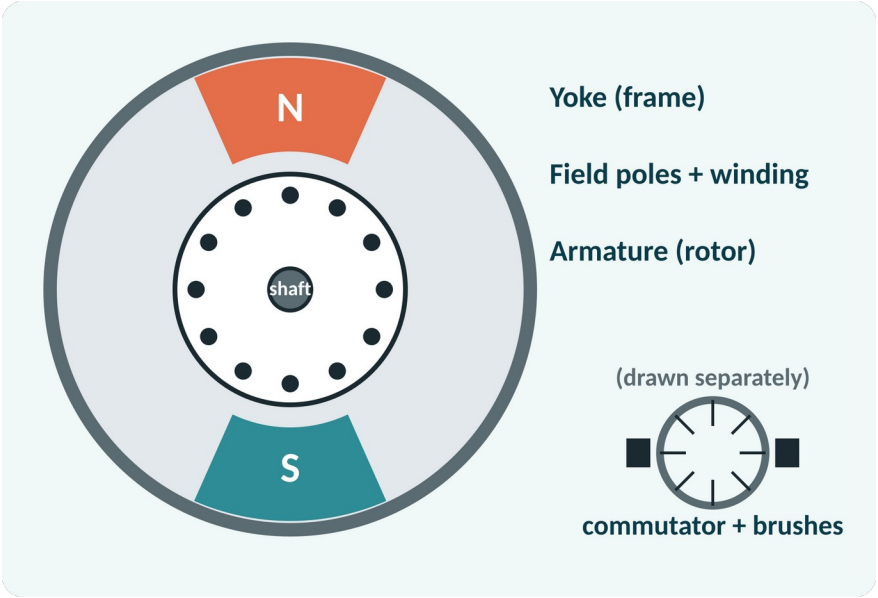
F = B·I·l (N)

Fleming left hand: thumb F, forefinger B, middle finger I

F = B·I·l (N); e = B·l·v (V)
B tesla, I ampere, l metre

Motivation: Where DC Motors Are Used

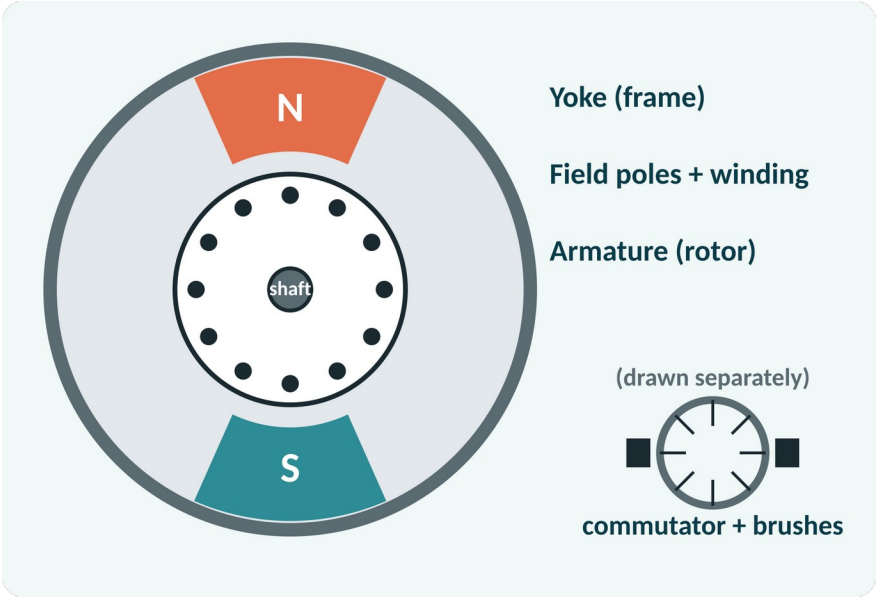
- Electric vehicles and trains.
- Cranes, lifts, rolling mills.
- Fans, toys, robotics, wipers.
- Easy and wide speed control.



DC motor: converts electrical power into mechanical power

Construction of a DC Motor

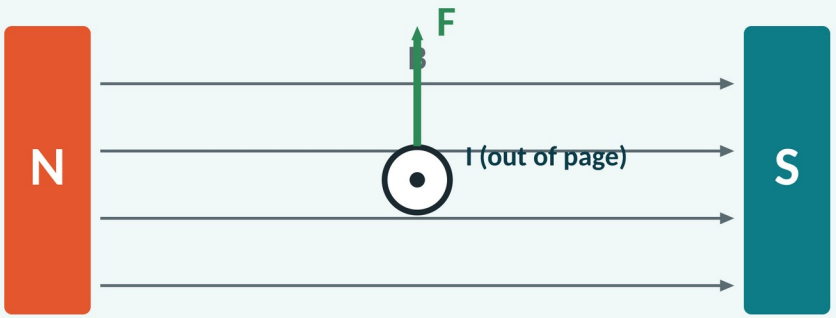
- Yoke: outer frame; carries flux.
- Field poles with field winding.
- Armature: laminated core with slots.
- Commutator: copper segments on the shaft.
- Brushes: carbon, supply current to armature.



Main parts: stator (yoke + poles), rotor (armature + commutator), brushes

Principle of Operation

- Armature conductors carry current in the field.
- Each conductor feels a force F = B·I·l.
- Forces on the two sides form a couple.
- The couple rotates the armature.
- Commutator keeps the torque unidirectional.



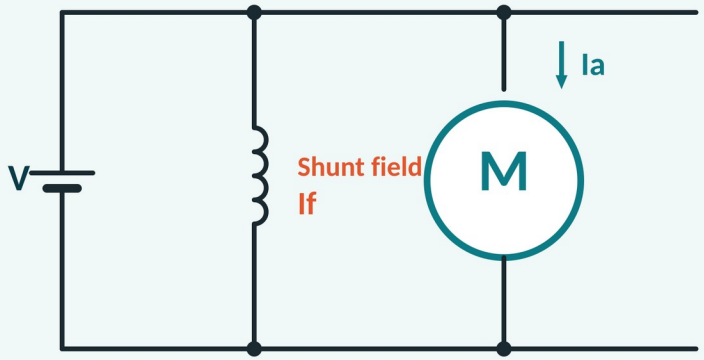
F = B·I·l (N)

Fleming left hand: thumb F, forefinger B, middle finger I

F = B·I·l; Torque T = F × r (N·m)

DC Shunt Motor Circuit

- Field winding connected across the supply.
- Field current If is nearly constant.
- Flux Φ is therefore constant.
- Armature draws Ia from the supply.
- Line current I = Ia + If.



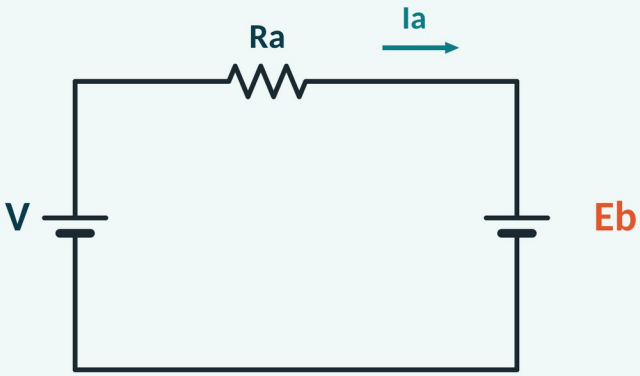
Field winding in parallel with the armature

I = Ia + If

I = Ia + If; If = V/Rsh

Back EMF and Its Significance

- Rotating conductors cut the field: EMF induced.
- This EMF opposes the supply: back EMF Eb.
- Ia = (V − Eb)/Ra.
- At start Eb = 0: very large current.
- Eb adjusts the current to the load: self-regulation.



V = Eb + Ia·Ra

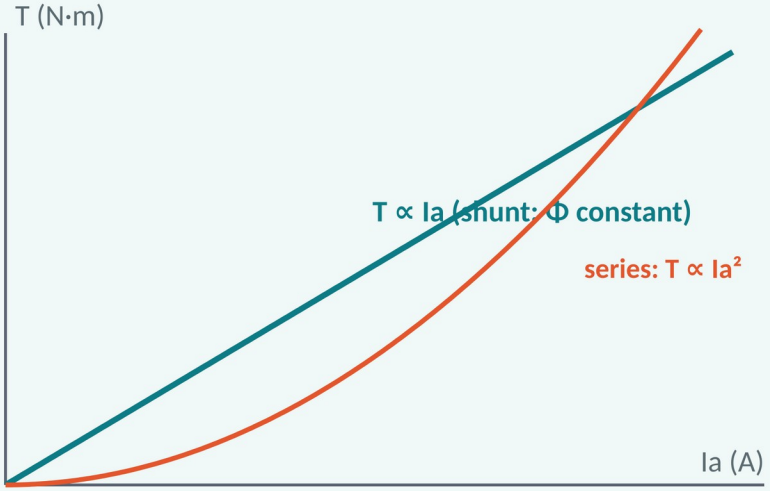
Eb opposes V; Ia = (V − Eb)/Ra

Eb = V − Ia·Ra = (P·Φ·Z·N)/(60·A); Pm = Eb·Ia

N speed rpm, Z conductors, P poles, A parallel paths

Torque Equation

- Torque comes from flux × armature current.
- Shunt motor: Φ constant, T ∝ Ia.
- Series motor: Φ ∝ Ia, T ∝ Ia^2.
- Mechanical power = Eb·Ia = ω·T.

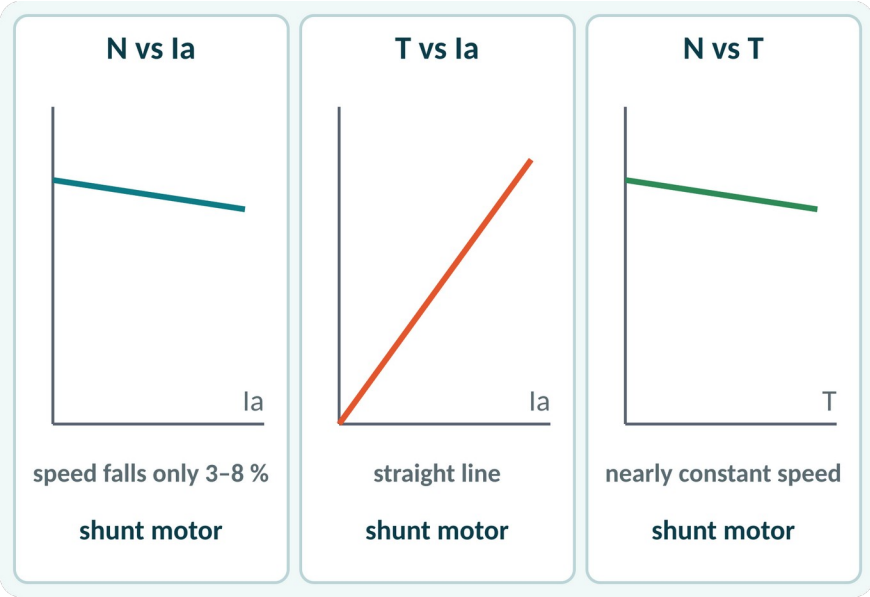


Pm = Eb·Ia = ω·T T = (PZ/2πA)·Φ·Ia = Kt·Φ·Ia

T = (P·Z/2πA)·Φ·Ia = Kt·Φ·Ia (N·m); Pm = Eb·Ia = 2πN·T/60
Φ weber, Ia ampere

Characteristics of a Shunt Motor

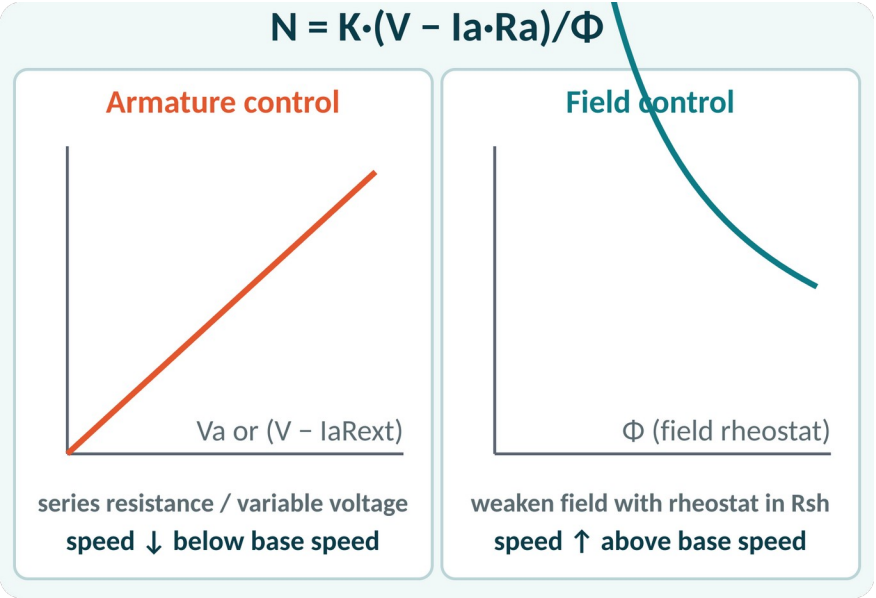
- Speed falls only slightly with load.
- Torque rises linearly with armature current.
- Practically constant-speed motor.
- Never run without field: speed runs away.



N ∝ Eb/Φ; T ∝ Ia (Φ constant)

Speed Control of a Shunt Motor

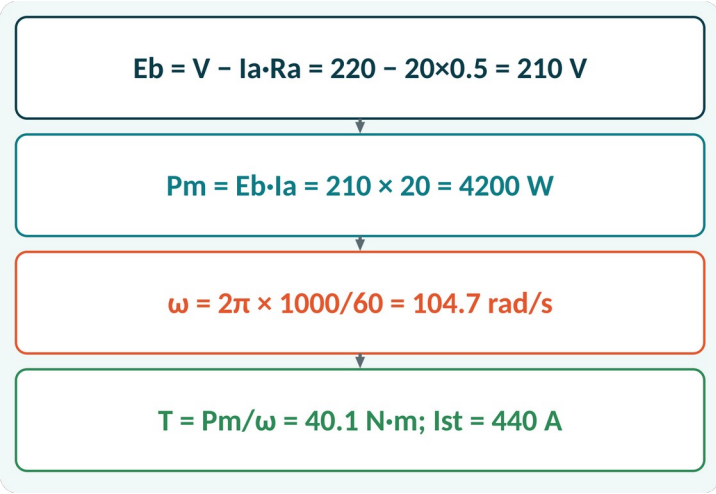
- Speed N ∝ (V − IaRa)/Φ.
- Armature control: vary voltage or add series R.
- Field control: weaken Φ with a field rheostat.
- Armature control gives speeds below base.
- Field control gives speeds above base.



N = K (V − Ia·Ra)/Φ; N2/N1 = (Eb2/Eb1)·(Φ1/Φ2)
K machine constant

Solved Example 1: Torque and Starting

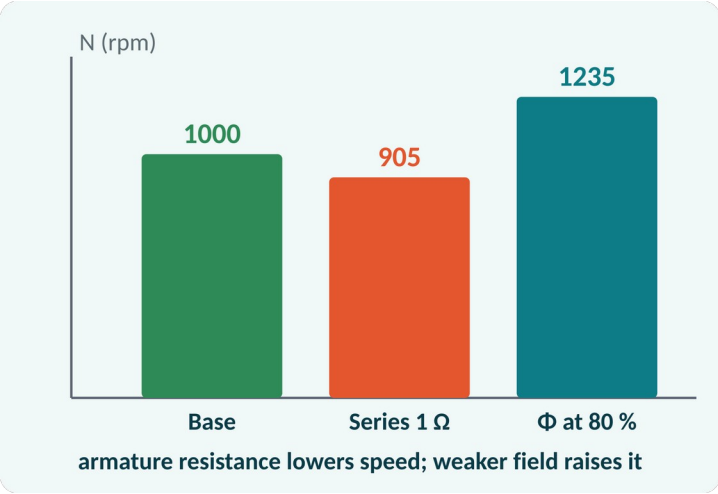
- A 220 V shunt motor, Ra = 0.5 Ω, runs at 1000 rpm taking Ia = 20 A. Find Eb, Pm, T and starting current without starter.
- 1. Eb = 220 − 20 × 0.5 = 210 V.
- 2. Pm = Eb·Ia = 4200 W.
- 3. ω = 2π × 1000/60 = 104.7 rad/s.
- 4. T = 4200/104.7 = 40.1 N·m.
- 5. Starting current = V/Ra = 220/0.5 = 440 A (22 × rated).



Answer: Eb = 210 V, Pm = 4.2 kW, T = 40.1 N·m, Ist = 440 A without a starter

Solved Example 2: Speed Control

- For the same motor with constant load torque, find the speed (a) with a 1 Ω resistance added in series with the armature, (b) with the field weakened to 80 %.
- (a) Ia = 20 A: Eb = 220 − 20 × 1.5 = 190 V; N = 1000 × 190/210 = 905 rpm.
- (b) T ∝ Φ·Ia: Ia = 20/0.8 = 25 A; Eb = 220 − 25 × 0.5 = 207.5 V.
- N = 1000 × (207.5/210) × (1/0.8) = 1235 rpm.



Answer: (a) 905 rpm, (b) 1235 rpm

Common Mistakes and Exam Tips

Common mistakes

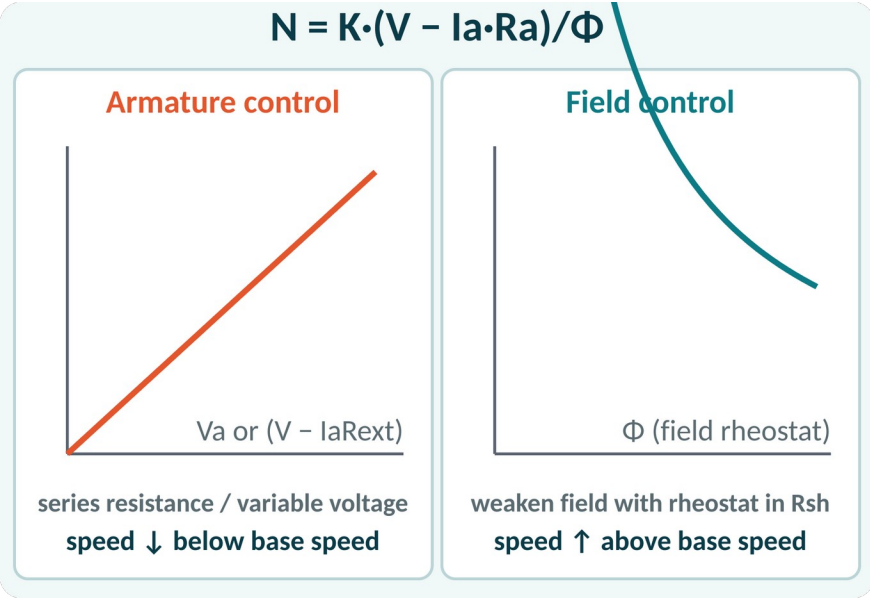
- Using V instead of Eb in P = Eb·Ia.
- Forgetting that Eb = 0 at start.
- Saying T ∝ Ia for a series motor.
- Opening the shunt field of a running motor.
- Mixing rpm with rad/s in T = P/ω.

Exam tips

- Compute Eb = V − IaRa first.
- Convert N to ω = 2πN/60.
- Use ratios for speed-change problems.
- Remember: field control raises speed.

Summary

- Motor action: force on a current-carrying conductor in a field.
- Back EMF: Eb = V − Ia·Ra opposes the supply.
- T = Kt·Φ·Ia; Pm = Eb·Ia.
- Shunt motor: nearly constant speed.
- Speed control: armature voltage (below), field flux (above).



N ∝ (V − IaRa)/Φ; T ∝ Φ·Ia

Quick Quiz

Q1. At the instant of starting, back EMF is:
(a) maximum (b) zero (c) equal to V (d) half of V

Ans: (b)

Q2. Torque of a shunt motor is proportional to:
(a) Ia^2 (b) Ia (c) 1/Ia (d) N

Ans: (b)

Q3. Field control gives speeds:
(a) below base (b) above base (c) zero (d) negative

Ans: (b)

Q4. V = 200 V, Ia = 10 A, Ra = 1 Ω gives Eb =
(a) 210 V (b) 190 V (c) 200 V (d) 10 V

Ans: (b)

Q5. The commutator in a motor:
(a) cools the machine (b) keeps torque unidirectional (c) increases flux (d) is a brake

Ans: (b)

References

1. **Hughes Edward, Electrical Technology**
   10th Ed., Pearson Education, 2010 (Units 1, 2 textbook)

2. **H. Cotton, Electrical Technology**
   7th Ed., CBS Publishers, 2005 (reference)

Next lecture: three-phase induction motor